

B1 Algebra — Lesson Plan

IGCSE Mathematics 0580 | Concept Guide

B1_Q1: Algebraic Expressions

Hook

Imagine you're a detective trying to figure out the cost of items without seeing the price tag. You see that buying 3 apples and 2 bananas costs \$5.50, while 1 apple and 4 bananas costs \$4.00. Algebra gives you the power to describe these relationships symbolically and find the hidden prices — like a mathematical superpower.

Prerequisites

- Basic arithmetic with integers and fractions
- Understanding of variables as placeholders for unknown numbers
- Order of operations (BIDMAS/BODMAS)

Core Explanation

Algebraic expressions are combinations of numbers, variables, and operations that represent mathematical relationships.

Key Vocabulary:

- **Variable**: A letter representing an unknown value (e.g., x , y , n)
- **Term**: A product of a coefficient and a variable (e.g., $3x$, $-2y^2$, 5)
- **Coefficient**: The numerical factor of a term (in $4x$, the coefficient is 4)
- **Constant**: A term with no variable (e.g., 7 , -3)
- **Like terms**: Terms with the same variable raised to the same power (e.g., $3x$ and $-5x$)

Collecting Like Terms:

To simplify an expression, group terms with the same variable and power:

$$3x + 5y - 2x + 7y = (3x - 2x) + (5y + 7y) = x + 12y$$

Expanding Brackets:

Use the distributive law: $a(b + c) = ab + ac$

$$3(2x + 5) = 3 \times 2x + 3 \times 5 = 6x + 15$$

For two brackets:

$$(x + 2)(x + 5) = x(x + 5) + 2(x + 5) = x^2 + 5x + 2x + 10 = x^2 + 7x + 10$$

The FOIL Method:

To multiply two binomials: First, Outer, Inner, Last

$$(x + 3)(x + 7) = x^2 + 7x + 3x + 21 = x^2 + 10x + 21$$

Squares of Binomials:

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$(a + b)(a - b) = a^2 - b^2$$

Worked Example

Problem: Simplify $3(2x - 4) - 2(x + 3) + 5x$

Step 1: Expand the brackets using the distributive law.

$$3(2x - 4) = 6x - 12$$

$$-2(x + 3) = -2x - 6$$

Step 2: Rewrite the full expression.

$$6x - 12 - 2x - 6 + 5x$$

Step 3: Group like terms.

$$x \text{ terms: } 6x - 2x + 5x = (6 - 2 + 5)x = 9x$$

$$\text{Constant terms: } -12 - 6 = -18$$

Step 4: Write the final simplified expression.

$$9x - 18$$

Check: Substitute $x = 1$ into the original:

$$3(2 - 4) - 2(1 + 3) + 5(1) = 3(-2) - 2(4) + 5 = -6 - 8 + 5 = -9$$

Substitute $x = 1$ into the simplified:

$$9(1) - 18 = 9 - 18 = -9 \checkmark$$

Common Mistakes

X Adding coefficients of unlike terms (e.g., $3x + 5 = 8x$)

OK Fix: You can only add terms that have the same variable and exponent. $3x$ and 5 are not like terms —

X Sign errors when expanding with a negative outside brackets (e.g., $-2(x - 3) = -2x - 6$)

OK Fix: The negative sign multiplies every term inside: $-2(x - 3) = -2 \times x + (-2) \times (-3) = -2x + 6$

X Forgetting to multiply every term inside the bracket (e.g., $3(2x + 5) = 6x + 5$)

OK Fix: The number outside multiplies EVERY term inside: $3(2x + 5) = 3 \times 2x + 3 \times 5 = 6x + 15$

Summary

- Like terms have the same variable and same exponent — only these can be combined by adding/subtracting coefficients
- The distributive law $a(b + c) = ab + ac$ is the foundation for expanding brackets
- When multiplying two binomials, multiply each term in the first by each term in the second (FOIL method)
- Always check sign carefully when a negative number sits outside brackets
- Simplify by expanding first, then collecting like terms

B1_Q2: Factorisation

Hook

Factorisation is like reverse multiplication — imagine watching a movie in rewind. Just as expanding brackets creates an expression, factorisation undoes that process to reveal the original building blocks. It's the key skill that unlocks solving quadratic equations and simplifying algebraic fractions.

Prerequisites

- Simplifying algebraic expressions and expanding brackets
- Multiplication tables and finding common factors
- Difference of two squares identity $(a+b)(a-b) = a^2 - b^2$

Core Explanation

Factorisation is the process of writing an expression as a product of its factors. It's the reverse of expanding brackets.

1. Common Factor Factorisation:

Find the highest common factor (HCF) of all terms and take it outside brackets.

$$6x^2 + 9x = 3x(2x + 3)$$

The HCF of $6x^2$ and $9x$ is $3x$.

2. Difference of Two Squares:

$$a^2 - b^2 = (a + b)(a - b)$$

This works for any expression that is one perfect square minus another.

$$x^2 - 25 = (x + 5)(x - 5) \quad 4x^2 - 9y^2 = (2x)^2 - (3y)^2 = (2x + 3y)(2x - 3y)$$

3. Factorising Quadratics ($x^2 + bx + c$):

Find two numbers p and q such that:

- $p + q = b$ (sum equals the coefficient of x)
- $p \times q = c$ (product equals the constant term)

$$\text{Then: } x^2 + bx + c = (x + p)(x + q)$$

Example: $x^2 + 7x + 12$

- Numbers that multiply to 12: (1,12), (2,6), (3,4)
- Numbers that add to 7: $3 + 4 = 7$
- Therefore: $x^2 + 7x + 12 = (x + 3)(x + 4)$

4. Factorising Quadratics ($ax^2 + bx + c$ where $a > 1$):

Use the 'ac method':

- Multiply a and c : $a \times c$
- Find two numbers that multiply to ac and add to b
- Rewrite the middle term as a sum of these two numbers
- Factor by grouping

Example: $2x^2 + 7x + 3$

- $a \times c = 2 \times 3 = 6$

- Numbers: $1 \times 6 = 6$ and $1 + 6 = 7 \checkmark$

- Rewrite: $2x^2 + 1x + 6x + 3$

- Group: $(2x^2 + x) + (6x + 3) = x(2x + 1) + 3(2x + 1) = (2x + 1)(x + 3)$

Worked Example

Problem: Factorise $3x^2 - 5x - 2$ completely.

Step 1: Identify $a=3$, $b=-5$, $c=-2$.

Step 2: Multiply a and c : $3 \times (-2) = -6$

Step 3: Find two numbers that multiply to -6 and add to -5 .

Factors of -6 : $(-6, 1)$, $(6, -1)$, $(-3, 2)$, $(3, -2)$

Check sums: $-6 + 1 = -5 \checkmark$

So the numbers are -6 and 1 .

Step 4: Rewrite the middle term.

$$3x^2 - 6x + 1x - 2$$

Step 5: Factor by grouping.

$$3x^2 - 6x + x - 2 = 3x(x - 2) + 1(x - 2) = (x - 2)(3x + 1)$$

Answer: $3x^2 - 5x - 2 = (x - 2)(3x + 1)$

Check: Expand: $(x - 2)(3x + 1) = 3x^2 + x - 6x - 2 = 3x^2 - 5x - 2 \checkmark$

Common Mistakes

X Forgetting to check if there is a common factor first (e.g., trying to factor $4x^2 - 16$ as difference of two squares)

OK Fix: Always take out the HCF first! $4x^2 - 16 = 4(x^2 - 4) = 4(x + 2)(x - 2)$

X Getting signs wrong when c is negative in $x^2 + bx + c$

OK Fix: If c is negative, one factor has a plus and the other has a minus. If c is positive, both factors have the same sign.

X Thinking $x^2 + 9$ can be factorised as $(x + 3)(x - 3)$

OK Fix: The difference of two squares formula requires a MINUS sign: $a^2 - b^2$. $x^2 + 9$ cannot be factorised.

Summary

- Always take out the highest common factor first before any other factorisation method
- Difference of two squares: $a^2 - b^2 = (a + b)(a - b)$
- Simple quadratics $x^2 + bx + c$: find two numbers with sum b and product c
- Harder quadratics $ax^2 + bx + c$: use the ac method — multiply a and c , split the middle

term, factor by grouping

- Check your answer by expanding the brackets back out

B1_Q3: Linear Equations

Hook

A set of balance scales is one of the oldest and most intuitive tools for understanding equations. When two sides balance, whatever you do to one side, you must do to the other. This simple idea — maintaining equality — is the foundation of solving every equation you'll ever encounter, from simple linear equations to complex differential equations.

Prerequisites

- Simplifying algebraic expressions
- Basic arithmetic operations with negatives and fractions
- Understanding of inverse operations

Core Explanation

A linear equation is an equation where the variable appears only to the first power (no x^2 , x^3 , etc.). The general form is $ax + b = c$ where $a \neq 0$.

The Golden Rule: Whatever you do to one side, you MUST do to the other.

Solving $ax + b = c$:

1. Subtract b from both sides: $ax = c - b$
2. Divide both sides by a : $x = \frac{c - b}{a}$

Example: Solve $3x + 7 = 22$

- Subtract 7: $3x = 15$
- Divide by 3: $x = 5$

Unknowns on Both Sides:

When the variable appears on both sides, collect all variable terms on one side and constants on the other.

Example: Solve $5x - 3 = 2x + 9$

- Subtract $2x$ from both sides: $3x - 3 = 9$
- Add 3 to both sides: $3x = 12$
- Divide by 3: $x = 4$

Equations with Brackets:

Always expand brackets first.

Example: Solve $2(3x - 1) = 4x + 10$

- Expand: $6x - 2 = 4x + 10$
- Subtract $4x$: $2x - 2 = 10$
- Add 2 : $2x = 12$
- Divide: $x = 6$

****Solving Linear Inequalities:****

Inequalities ($<$, $>$, $\leq$, $\geq$) are solved the same way as equations, with one critical exception:

> **If you multiply or divide by a negative number, flip the inequality sign.**

Example: Solve $-3x + 5 \leq 14$

- Subtract 5 : $-3x \leq 9$
- Divide by -3 (FLIP sign): $x \geq -3$

The solution can be shown on a number line.

****Solving Fractional Equations:****

Multiply both sides by the common denominator to eliminate fractions.

Example: Solve $\frac{x}{3} + \frac{1}{2} = \frac{5}{6}$

- Multiply by 6 (LCM of $3, 2, 6$): $2x + 3 = 5$
- Subtract 3 : $2x = 2$
- Divide: $x = 1$

Worked Example

****Problem:**** Solve $\frac{2x - 1}{3} - \frac{x + 2}{4} = 2$

****Step 1:**** Find the LCM of denominators. LCM of 3 and 4 is 12 .

****Step 2:**** Multiply both sides by 12 .

$$12 \times \left(\frac{2x - 1}{3} - \frac{x + 2}{4} \right) = 12 \times 2$$

****Step 3:**** Distribute the 12 to each fraction.

$$\frac{12}{3}(2x - 1) - \frac{12}{4}(x + 2) = 24$$

$$4(2x - 1) - 3(x + 2) = 24$$

****Step 4:**** Expand brackets.

$$8x - 4 - 3x - 6 = 24$$

****Step 5:**** Collect like terms.

$$5x - 10 = 24$$

****Step 6:**** Solve for x .

$$5x = 34$$

$$x = \frac{34}{5} = 6.8$$

****Check:**** $\frac{2(6.8) - 1}{3} - \frac{6.8 + 2}{4} = \frac{13.6 - 1}{3} - \frac{8.8}{4} =$

$$\frac{12.6}{3} - 2.2 = 4.2 - 2.2 = 2 \checkmark$$

Common Mistakes

X Forgetting to flip the inequality sign when multiplying or dividing by a negative number

OK Fix: Remember: multiplying or dividing by a negative reverses the inequality direction. For example, $-2x < 5$ becomes $x > -2.5$.

X Only multiplying one term when clearing fractions (e.g., $\frac{x}{2} + 3 = 5$ becomes $x + 3 = 5$)

OK Fix: Multiply EVERY term on both sides by the denominator: $2 \times \frac{x}{2} + 2 \times 3 = 2 \times 5$

X Sign errors when moving terms across the equals sign

OK Fix: Instead of 'moving terms', think of doing the same operation to both sides. To remove a $+5$ on the left, subtract 5 from both sides.

Summary

- The golden rule: do the same operation to both sides of the equation
- Use inverse operations in reverse order of BIDMAS to isolate the variable
- For inequalities, flip the sign when multiplying or dividing by a negative number
- Clear fractions by multiplying both sides by the LCM of all denominators
- Always check your answer by substituting back into the original equation

B1_Q4: Simultaneous Equations

Hook

Imagine you're at a market stall with two identical-looking bags of fruit. One bag has 3 apples and 2 bananas costing \$2.80. Another has 1 apple and 4 bananas costing \$2.60. Can you figure out the individual prices? This is exactly what simultaneous equations do — they use two pieces of information to find two unknown values.

Prerequisites

- Solving linear equations
- Manipulating algebraic expressions
- Understanding of coordinates and graphs

Core Explanation

Simultaneous equations are a system of two (or more) equations that share the same variables. The solution is the pair(s) of values that satisfy all equations at once.

****Method 1: Elimination****

Make the coefficients of one variable the same in both equations, then add or subtract to eliminate that variable.

Example: Solve $3x + y = 11$ and $2x - y = 4$

1. The y coefficients are +1 and -1. Adding eliminates y.
2. $(3x + y) + (2x - y) = 11 + 4$
3. $5x = 15$, so $x = 3$
4. Substitute back: $3(3) + y = 11$, so $9 + y = 11$, $y = 2$
5. Solution: $x = 3$, $y = 2$

****Method 2: Substitution****

Make one variable the subject of one equation, then substitute into the other.

Example: Solve $y = 2x + 1$ and $3x + 2y = 16$

1. Substitute $y = 2x + 1$ into the second equation:
2. $3x + 2(2x + 1) = 16$
3. $3x + 4x + 2 = 16$
4. $7x = 14$, so $x = 2$
5. Substitute back: $y = 2(2) + 1 = 5$
6. Solution: $x = 2$, $y = 5$

****Choosing the Method:****

- Elimination is best when coefficients are small and cancellation is easy
- Substitution is best when one equation already has a variable isolated (e.g., $y = \dots$ or $x = \dots$)

****Graphical Method:****

Each equation represents a straight line. The solution is the point where the two lines intersect. Plot both lines; the intersection coordinates are the solution.

****Solving Word Problems:****

1. Define variables (x and y) for the unknown quantities
2. Write two equations based on the information given
3. Solve using elimination or substitution
4. Check the answer makes sense in the original context

Worked Example

****Problem:**** Solve $2x + 3y = 7$ and $5x - 2y = 8$

****Step 1:**** Choose elimination. The LCM of 3 and 2 is 6. Multiply to match y coefficients.

- Equation 1 $\times 2$: $4x + 6y = 14$
- Equation 2 $\times 3$: $15x - 6y = 24$

****Step 2:**** Add the equations to eliminate y.

$$(4x + 6y) + (15x - 6y) = 14 + 24$$
$$19x = 38$$
$$x = 2$$

****Step 3:**** Substitute $x = 2$ into Equation 1.

$$2(2) + 3y = 7$$
$$4 + 3y = 7$$

$$3y = 3$$

$$y = 1$$

Step 4: Check in Equation 2.

$$5(2) - 2(1) = 10 - 2 = 8 \checkmark$$

Answer: $x = 2, y = 1$

Common Mistakes

X Adding when you should subtract (or vice versa) when eliminating variables

OK Fix: If the coefficients have the SAME sign, subtract. If they have OPPOSITE signs, add. Write this out clearly

X Making sign errors when substituting a negative value back into an equation

OK Fix: Use brackets when substituting: if $x = -2$, write $3(-2) + 2y = 5$ not $3(-2) + 2y = 5$ — actually the

X Forgetting to check the solution in BOTH equations

OK Fix: A solution that works in one equation but not the other is NOT a valid solution to the system. Always ve

Summary

- Elimination: match coefficients of one variable, then add or subtract to eliminate it
- Substitution: isolate one variable and substitute into the other equation
- Graphically: the solution is the intersection point of the two lines
- Always check your answer in both original equations
- Word problems: define variables, form equations, solve, then interpret results

B1_Q5: Algebraic Fractions

Hook

Fractions with numbers are straightforward enough — everyone knows $\frac{1}{3} + \frac{1}{6} = \frac{1}{2}$. But what happens when the numerators and denominators contain variables? Algebraic fractions are the bridge between numerical arithmetic and the abstract world of rational expressions, and mastering them is essential for calculus, physics, and engineering.

Prerequisites

- Numerical fraction arithmetic (adding, subtracting, multiplying, dividing)
- Factorisation techniques (common factors, quadratics, difference of squares)
- Solving linear equations

Core Explanation

Algebraic fractions follow the exact same rules as numerical fractions — the only difference is that the numbers are replaced by algebraic expressions.

****Simplifying Algebraic Fractions:****

Factorise numerator and denominator, then cancel common factors.

Example: Simplify $\frac{x^2 - 9}{x^2 - x - 6}$

- Factorise numerator: $x^2 - 9 = (x + 3)(x - 3)$

- Factorise denominator: $x^2 - x - 6 = (x + 2)(x - 3)$

- Cancel common factor $(x - 3)$:

$$\frac{(x + 3)(x - 3)}{(x + 2)(x - 3)} = \frac{x + 3}{x + 2}$$

****Multiplying Algebraic Fractions:****

Multiply numerators together and denominators together. Factorise first to simplify cancellation.

$$\frac{2x}{3} \times \frac{6}{x^2} = \frac{2x \times 6}{3 \times x^2} = \frac{12x}{3x^2} = \frac{4}{x}$$

****Dividing Algebraic Fractions:****

Flip the second fraction (reciprocal) and multiply.

$$\frac{x}{4} \div \frac{x^2}{2} = \frac{x}{4} \times \frac{2}{x^2} = \frac{2x}{4x^2} = \frac{1}{2x}$$

****Adding and Subtracting Algebraic Fractions:****

Find a common denominator (LCM of the denominators), rewrite each fraction, then add/subtract the numerators.

Example: $\frac{2}{x} + \frac{3}{x + 1}$

- Common denominator: $x(x + 1)$

- Rewrite: $\frac{2(x + 1)}{x(x + 1)} + \frac{3x}{x(x + 1)}$

- Add numerators: $\frac{2(x + 1) + 3x}{x(x + 1)} = \frac{2x + 2 + 3x}{x(x + 1)} = \frac{5x + 2}{x(x + 1)}$

****Solving Equations with Algebraic Fractions:****

Multiply both sides by the LCM of all denominators to clear fractions, then solve the resulting equation.

Example: Solve $\frac{3}{x} + \frac{1}{2} = \frac{5}{2x}$

- Multiply by $2x$: $6 + x = 5$

- Solve: $x = -1$

- Check: $\frac{3}{-1} + \frac{1}{2} = -3 + 0.5 = -2.5$ and $\frac{5}{2(-1)} = \frac{5}{-2} = -2.5$ ✓

Worked Example

****Problem:**** Simplify $\frac{1}{x - 2} - \frac{2}{x + 1} + \frac{3x - 7}{x^2 - x - 2}$

****Step 1:**** Factorise any quadratic denominators.

$$x^2 - x - 2 = (x - 2)(x + 1)$$

****Step 2:**** The LCM of $(x - 2)$, $(x + 1)$, and $(x - 2)(x + 1)$ is $(x - 2)(x + 1)$.

Step 3: Rewrite each fraction with the common denominator.

$$\frac{1(x+1)}{(x-2)(x+1)} - \frac{2(x-2)}{(x-2)(x+1)} + \frac{3x-7}{(x-2)(x+1)}$$

Step 4: Combine numerators.

$$\frac{(x+1) - 2(x-2) + (3x-7)}{(x-2)(x+1)}$$

Step 5: Simplify the numerator.

$$x+1-2x+4+3x-7 = (x-2x+3x) + (1+4-7) = 2x-2$$

Step 6: Factorise the numerator if possible.

$$\frac{2x-2}{(x-2)(x+1)} = \frac{2(x-1)}{(x-2)(x+1)}$$

Answer: $\frac{2(x-1)}{(x-2)(x+1)}$, $x \neq 2, -1$

Common Mistakes

X Cancelling terms instead of factors (e.g., $\frac{x+3}{x+5} = \frac{3}{5}$ by cancelling x)

OK Fix: You can only cancel FACTORS (things being multiplied), not terms (things being added). $x+3$ and $x+5$

X Forgetting to multiply every term when clearing fractions in an equation

OK Fix: When multiplying both sides by the LCM, multiply EVERY term, not just the fractions. Use brackets to be sure.

X Sign errors when subtracting a fraction (e.g., $\frac{2}{x} - \frac{x+1}{x+3} = \frac{2}{x} - \frac{x+1}{x+3}$)

OK Fix: When subtracting, put the ENTIRE numerator in brackets: $\frac{2}{x} - \frac{x+1}{x+3} = \frac{2}{x} - \frac{x+1}{x+3}$

Summary

- Algebraic fractions follow the same rules as numerical fractions
- Always factorise numerators and denominators before simplifying or adding
- To add/subtract, use the LCM of denominators as the common denominator
- To clear fractions from an equation, multiply both sides by the LCM
- State restrictions: values that make any denominator zero are not allowed

B1_Q6: Sequences

Hook

The ancient Greeks discovered that if you arrange pebbles in triangular patterns — 1, then 3, then 6, then 10 pebbles — each new triangle adds a row with one more pebble than the last. This pattern of triangular numbers was one of the first sequences ever studied. Today, sequences appear everywhere: in loan repayments, population growth, computer algorithms, and even in the spiral patterns of sunflowers.

Prerequisites

- Basic arithmetic with negative numbers

- Substituting values into algebraic expressions
- Solving simple linear equations

Core Explanation

A sequence is an ordered list of numbers following a rule. Each number in a sequence is called a term.

****Linear Sequences (Arithmetic Progressions):****

A sequence where the difference between consecutive terms is constant. The n th term is:

$$u_n = a + (n - 1)d$$

where a is the first term and d is the common difference.

Example: 3, 7, 11, 15, 19, ...

- First term $a = 3$, common difference $d = 4$
- n th term: $u_n = 3 + (n - 1)4 = 4n - 1$
- To find the 10th term: $u_{10} = 4(10) - 1 = 39$

****Finding the n th term of a linear sequence:****

The n th term has the form $an + b$ where a is the common difference and b adjusts for the first term.

1. Find d (the common difference) — this is the coefficient of n
2. Substitute $n = 1$ to find the adjustment: $a + b = \text{first term}$

****Quadratic Sequences:****

A sequence where the second difference (difference of differences) is constant. The n th term has the form:

$$u_n = an^2 + bn + c$$

To find a , b , c :

1. Find the first and second differences
2. $a = \frac{\text{second difference}}{2}$
3. Subtract an^2 from each term to get a linear sequence
4. Find the n th term of the remaining linear sequence

Example: 3, 6, 11, 18, 27, ...

- First differences: 3, 5, 7, 9
- Second difference: 2 (constant) $\rightarrow a = \frac{2}{2} = 1$
- Subtract n^2 : $3 - 1 = 2$, $6 - 4 = 2$, $11 - 9 = 2$, $18 - 16 = 2$, $27 - 25 = 2$
- Remaining sequence: 2, 2, 2, 2, 2 $\rightarrow$ constant 2
- So $u_n = n^2 + 2$

****Term-to-Term Rules:****

A recursive definition tells you how to get the next term from the current one.

- Linear: $u_{n+1} = u_n + d$
- Example: 3, 7, 11, 15, ... has term-to-term rule $u_{n+1} = u_n + 4$, $u_1 = 3$

****Special Sequences:****

- Triangular numbers: 1, 3, 6, 10, 15, ... $\rightarrow u_n = \frac{n(n+1)}{2}$

- Square numbers: 1, 4, 9, 16, 25, \dots $\rightarrow u_n = n^2$
- Fibonacci: 1, 1, 2, 3, 5, 8, \dots $\rightarrow u_{n+1} = u_n + u_{n-1}$

Worked Example

****Problem:**** Find the n th term of the sequence 5, 9, 15, 23, 33, \dots and use it to find the 20th term.

****Step 1:**** Check if linear or quadratic.

First differences: $9 - 5 = 4$, $15 - 9 = 6$, $23 - 15 = 8$, $33 - 23 = 10$

Second difference: $6 - 4 = 2$, $8 - 6 = 2$, $10 - 8 = 2 \rightarrow$ constant, so quadratic.

****Step 2:**** Find a .

$$a = \frac{\text{second difference}}{2} = \frac{2}{2} = 1$$

****Step 3:**** Subtract $an^2 = n^2$ from each term.

$$n = 1: 5 - 1^2 = 4$$

$$n = 2: 9 - 2^2 = 5$$

$$n = 3: 15 - 3^2 = 6$$

$$n = 4: 23 - 4^2 = 7$$

$$n = 5: 33 - 5^2 = 8$$

Remaining: 4, 5, 6, 7, 8, \dots

****Step 4:**** This is a linear sequence with $d = 1$, first term 4.

$$\text{Linear } n\text{th term: } 4 + (n - 1)(1) = n + 3$$

****Step 5:**** Combine with an^2 .

$$u_n = n^2 + n + 3$$

****Step 6:**** Find the 20th term.

$$u_{20} = 20^2 + 20 + 3 = 400 + 20 + 3 = 423$$

****Answer:**** $u_n = n^2 + n + 3$, and $u_{20} = 423$

Common Mistakes

X Using $a + nd$ instead of $a + (n - 1)d$ for linear sequences

OK Fix: When $n = 1$, we need $u_1 = a$. If you use $a + nd$, $u_1 = a + d$ which is wrong. The correct formula is $a + (n - 1)d$.

X Confusing first difference with common difference — saying $d = 4$ when first differences are 4, 6, 8, 10, ...

OK Fix: For a linear sequence, ALL first differences must be equal. If they're not equal, check for a quadratic sequence.

X Writing the n th term of a quadratic without checking it works for $n = 1$

OK Fix: Always verify: substitute $n = 1, 2, 3$ into your n th term formula and check you get the original sequence.

Summary

- Linear sequence n th term: $u_n = a + (n - 1)d$ where a is first term, d is common difference

- For quadratic sequences, find the second difference; $a = \frac{\text{second difference}}{2}$
- To find quadratic n th term: subtract an^2 , find linear n th term of remainder, then add an^2
- Term-to-term rules describe how to get the next term from the current one
- Always verify your formula by checking the first few terms

B1_Q7: Functions and Graphs

Hook

Every time you use a vending machine, you're experiencing a function: you press a button (input), the machine processes it, and a drink comes out (output). A function is simply a rule that takes an input and produces exactly one output. In mathematics, we use functions to model everything from the trajectory of a rocket to the growth of a savings account.

Prerequisites

- Substituting values into algebraic expressions
- Plotting points on a coordinate grid
- Basic understanding of algebraic notation

Core Explanation

Function Notation:

A function is a rule that maps each input to exactly one output. We write $f(x)$ (read 'f of x') to represent the output when the input is x .

$$f(x) = 3x + 5$$

$$\text{- } f(2) = 3(2) + 5 = 11 \text{ (when the input is 2, the output is 11)}$$

$$\text{- } f(0) = 3(0) + 5 = 5$$

$$\text{- } f(-1) = 3(-1) + 5 = 2$$

Domain and Range:

- **Domain**: all possible input values

- **Range**: all possible output values

For $f(x) = x^2$ with domain $\{-2, -1, 0, 1, 2\}$:

- Range: $\{0, 1, 4\}$

For $f(x) = 2x + 1$ with domain $\mathbb{R}$ (all real numbers):

- Range: $\mathbb{R}$ (all real numbers)

Linear Graphs:

The graph of $y = mx + c$ is a straight line where:

- m is the **gradient** (steepness): $m = \frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x}$

$y \pm \Delta x$

- c is the **y-intercept** (where the line crosses the y-axis)

Finding the Gradient:

Given two points (x_1, y_1) and (x_2, y_2) on a line:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Example: The line through $(1, 3)$ and $(4, 9)$:

$$m = \frac{9 - 3}{4 - 1} = \frac{6}{3} = 2$$

Equation of a Line:

If you know the gradient m and a point (x_1, y_1) :

$$y - y_1 = m(x - x_1)$$

Example: Line through $(2, 5)$ with gradient 3:

$$y - 5 = 3(x - 2)$$

$$y - 5 = 3x - 6$$

$$y = 3x - 1$$

Parallel and Perpendicular Lines:

- Parallel lines have equal gradients: $m_1 = m_2$

- Perpendicular lines: $m_1 \times m_2 = -1$ (negative reciprocal)

So if line A has gradient 2, a perpendicular line has gradient $-\frac{1}{2}$.

Sketching Linear Graphs:

To sketch $y = mx + c$:

1. Plot the y-intercept $(0, c)$
2. Use the gradient: from the intercept, move right 1 unit and up/down by m units
3. Draw the line through these points

Worked Example

Problem: A line passes through $(3, -2)$ and $(5, 6)$. Find its equation. Then find the equation of a line perpendicular to it passing through $(3, -2)$.

Step 1: Find the gradient.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6 - (-2)}{5 - 3} = \frac{8}{2} = 4$$

Step 2: Use point-slope form with $(3, -2)$.

$$y - (-2) = 4(x - 3)$$

$$y + 2 = 4x - 12$$

$$y = 4x - 14$$

Step 3: Find perpendicular gradient.

$$m_{\text{perp}} = -\frac{1}{4} \text{ (negative reciprocal of 4)}$$

****Step 4:**** Equation of perpendicular line through (3, -2).

$$y - (-2) = -\frac{1}{4}(x - 3)$$

$$y + 2 = -\frac{1}{4}x + \frac{3}{4}$$

$$y = -\frac{1}{4}x + \frac{3}{4} - 2$$

$$y = -\frac{1}{4}x - \frac{5}{4}$$

****Answer:****

First line: $y = 4x - 14$

Perpendicular line: $y = -\frac{1}{4}x - \frac{5}{4}$

Common Mistakes

X Confusing $f(2)$ with $2f(x)$ — $f(2)$ means substitute $x = 2$, but $2f(x)$ means multiply the

OK Fix: $f(2)$ evaluates the function at $x = 2$: if $f(x) = x + 3$, then $f(2) = 5$. But $2f(x) = 2(x + 3) = 2x + 6$

X Reversing the gradient formula: $(y_2 - y_1)/(x_2 - x_1)$ reversed

OK Fix: Gradient is $\frac{\text{change in } y}{\text{change in } x}$. Always put y -difference on top, x -difference on bottom

X Saying perpendicular lines have gradients that multiply to 1 instead of -1

OK Fix: Perpendicular lines satisfy $m_1 \times m_2 = -1$. For example, 2 and $-\frac{1}{2}$ are perpendicular

Summary

- Function notation $f(x)$ means 'the output when the input is x '
- Domain = set of inputs, Range = set of outputs
- Linear graphs: $y = mx + c$, where m is gradient and c is y -intercept
- Gradient $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}$
- Parallel lines have equal gradients; perpendicular lines have negative reciprocal gradients ($m_1 \times m_2 = -1$)
- Point-slope form: $y - y_1 = m(x - x_1)$